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Tracking *Plasmodium falciparum* antimalarial resistance markers during a malaria pre-elimination period in the Pacific coast of South America

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Antimalarial resistance in *Plasmodium falciparum* is a public health problem in the fight against malaria in Ecuador. Characterizing the molecular epidemiology of drug resistance genes helps to understand the emergence and spread of resistant parasites. In this study, the effects of drug pressure and human migration on antimalarial resistance in *P. falciparum* were evaluated. Sixty-seven samples from northwestern Ecuador from the 2019–2021 period were analyzed. SNPs in *Pfcrt*, *Pfdhps*, *Pfdhfr*, *Pfmdr-1*, *Pfk13* and *Pfaat1* were identified by Sanger sequencing and whole-genome sequencing. A comparison of the frequencies of the haplotypes was made with data from the 2013–2015 period. Also, nucleotide and haplotype diversity were calculated. The frequencies of the mutant haplotypes, CVMET in *Pfcrt* and CICNI in *Pfdhfr* increased and became dominant (100% of infections) in Esmeraldas. NEDFSDFY in *Pfmdr-1* was detected for the first time, while two wild-type haplotypes, SAKAA in *Pfdhps* and MYRIC in *Pfk13*, remained in 100% of samples. Interestingly, the A16V mutation in *Pfdhfr* that gives resistance to proguanil is reported in Ecuador for the first time. In conclusion, parasites resistant to chloroquine (*Pfcrt*) and pyrimethamine (*Pfdhfr*) increased in recent years, while parasites sensitive to sulfadoxine (*Pfdhps*) and artemisinin (*Pfk13*) prevail in Ecuador. These results suggest that the current first-line treatment (artemether-lumefantrine + primaquine) is still useful against *P. falciparum*.

Keywords *Plasmodium falciparum*, Resistance, Antimalarial, Selective pressure, Human migration

Malaria is a vector-borne infectious disease endemic in tropical and subtropical regions of the world[1]. *Plasmodium falciparum* causes the most severe malaria in humans and the highest morbidity and mortality worldwide[2]. According to the World Health Organization (WHO), in 2022, 249 million cases and 608,000 deaths by malaria occurred in the world, and 600,000 cases and an incidence of 3.6 cases per 1000 inhabitants occurred in the Americas[3]. In this region, the highest malaria incidence is attributed to *P. vivax* followed by *P. falciparum*[1,4].

Since the approval of artemisinin-based combination therapy (ACT) against *P. falciparum*, several South American countries have implemented it in their treatment schemes[5,6]. Artemether (AR) and artesunate (AS) are the most common artemisinins (ARTs) used in ACTs[5]. Currently, seven of eleven countries in this region, including Ecuador, use artemether-lumefantrine (AL) + primaquine (PQ) as first-line treatment and quinine (QN) + clindamycin (CD) as second-line treatment[5,7].

Antimalarial resistance in *P. falciparum* is a public health problem that complicates malaria control in endemic regions[8,9]. Resistance is an evolutionary response to pressure exerted by drugs over parasites that allows them to survive and reproduce (fitness) during exposure to antimalarials[6]. This advantage is due to mutations that modify protein function and reduce antimalarial efficacy[10,11]. In *P. falciparum*, Single Nucleotide Polymorphisms (SNPs) have been identified in specific genes that are related to resistance to antimalarials such as chloroquine (CQ), sulfadoxine-pyrimethamine (SP), and ARTs, among others[12–14].

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CQ resistance in *P. falciparum* originated in Southeast Asia in the late 1950s[14]. Worldwide, *P. falciparum* exhibits CQ resistance mediated by mutations in *Pfcrt* that change the configuration of the chloroquine resistance transporter (PfCRT), located in the digestive vacuole (DV), allowing drug expulsion[15,16]. High concentrations of CQ in the DV prevent the release of toxic endogenous products, which cause the death of the parasite[17]. Mutations at codons 72–76 are associated with CQ resistance[18,19]. In vitro studies show that all chloroquine-resistant parasites have the K76**T** mutation, essential to confer CQ resistance[15,19]. In the early 1960s, CQ resistance emerged independently in South America[14], and CVMN**T**, CVM**ET** and CV**EIT** haplotypes have been found to predominate on the Pacific coast and **S**VMN**T** in the Amazon region[20,21]. Specifically, in Ecuador, CVMN**T**, CVM**ET** and **S**VMN**T** have been previously reported[20,22]. In addition, other mutations such as H97**Q**, A220**S**, Q271**E**, N326**D**, C350**R**, I356T/**L** and R371**I** in *Pfcrt* contribute to different degrees of CQ resistance[15,23,24]. In South America, H97**Q**, A220**S**, N326**D**, S334**N**, I356**L** and R371**I**/**T** have been found[16,17,23,25]. While in Ecuador, A220**S**, N326**D** and I356**L** with CVMN**T** were identified in Ecu1110, a CQ resistant isolate from 1990[16,20].

Resistance to the SP drug combination in *P. falciparum* originated in Southeast Asia in the 1960s[14]. In South America, SP treatment failure has been registered since the 1980s[26]. SP resistance is associated with mutations in *Pfdhps* and *Pfdhfr*, which encode synergistic enzymes: dihydropteroate synthetase (DHPS) and dihydrofolate reductase (DHFR), respectively[13,27]. Both enzymes participate in the folate synthetic pathway, essential for DNA synthesis and the metabolism of some amino acids[29]. The presence of SP causes competitive inhibition of the active site of DHPS and DHFR, causing parasite death[28,29]. Mutations at codons 436, 437, 540, 581 and 613 in *Pfdhps* confer sulfadoxine resistance. In South America, SAKAA (wild type), **F**AKAA, **FG**KAA, S**G**KAA, SA**E**AA, S**GK**A and S**GEG**A have been reported[26,30–33], while only SAKAA and SA**E**AA have been found in Ecuador[20]. On the other hand, mutations at codons 50, 51, 59, 108 and 164 in *Pfdhfr* confer pyrimethamine resistance[34]. In vitro studies show that S108**N** is the first mutation to emerge in *Pfdhfr*, which is essential to cause pyrimethamine resistance[34,35]. In South America, CNCSI (wild-type), CNC**N**I, C**I**C**N**I, C**I**C**NL**, CNC**T**I and **RI**C**N**I haplotypes have been reported[26,31,33], while CNCSI, CNC**N**I and C**I**C**N**I have been found in Ecuador[20].

Mutations in *Pfmdr-1* as well as its copy number are associated with resistance to several drugs such as CQ, QN, amodiaquine (AQ), mefloquine (MQ) and lumefantrine (LF)[15,36]. Mutations at codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246 change the charge and structure of the multidrug resistance transporter (PfMDR1), located in the DV, causing drug expulsion[37,38]. N86**Y**, Y184**F**, S1034**C**, N1042**D** and D1246**Y** mutations have been reported in South America[24,36], while Y184**F** and N1042**D** have been found in Ecuador[20].

ART resistance in *P. falciparum* was first reported in Southeast Asia in 2006[39]. Initially, ARTs are activated by heme and iron to then cause protein and proteasome damage, leading to prolonged cellular stress and parasite death[40]. Mutations at codons 476, 493, 539, 543 and 580 in *Pfk13* are associated with ART resistance[36,40,41]. In South America, C580**Y** has only been found in Guyana since 2010[41,42], while Ecuador, Colombia, Peru, Venezuela, Brazil, Suriname and French Guiana have not reported any mutation associated with ART resistance[20,42,43]. Similarly, neither Ecuador nor any of its neighboring countries have reported any instances of clinical treatment failure with ACTs.

In recent years, amino acid transporter 1 (*Pfaat1*) has been associated with resistance to quinolines such as QN, AQ MQ and especially CQ[44–46.] A study suggests that the PfAAT1 protein, located in the DV, transports amino acids out of the DV while allowing entry of quinolines[46,47]. However, the presence of mutations such as T162**E** changes the charge of PfAAT1, blocking drug entry into the DV, and causes resistance to CQ, QN, AQ and MQ[46]. One PfAAT1 mutation with demonstrated CQ-resistance phenotype (S258**L**) was found on the Pacific coast of Ecuador and Colombia[48,49].

In South America, *P. falciparum* populations exhibit varying degrees of enhanced relatedness as a result of bottlenecks caused by efforts to eliminate malaria in the region[11,21,50]. In Esmeraldas, Ecuador, a malaria outbreak was caused by a clonal expansion in 2013[21]. In addition, malaria outbreaks are influenced by the spread of parasites through human hosts between endemic regions[21,22,51]. In the malaria-endemic countries of the Americas, 7775 imported cases were reported in 2016 and 8411 in 2017, while in Ecuador, the latest records reported 395 cases between 2015 and 2017, with 41.5% of them due to *P. falciparum*[5,52].

In 2012, Ecuador was considered one of 21 countries in the world with the potential to eliminate malaria by 2020 due to a reduction in cases since 2002 of more than 99%[2]. The decrease in malaria infections was largely due to the success of the malaria elimination programs; however, infections have increased since 2015[4]. This increase may be associated with factors such as the restructuring of the health system, climate change, indiscriminate drug use and the spread of parasites by vector and host[53,54]. Genetic surveillance can help to resolve whether the emergence of new resistant genotypes in *P. falciparum* is the result of de novo mutations or the spread of parasites by human host migration.

This molecular epidemiological study of *P. falciparum* describes the frequency of previously characterized variants in drug-resistance genes that have been subjected to the combined effects of regional drug pressure and human migration. This work aims to characterize the current epidemiological situation of resistance markers and contribute to the strengthening of health programs by predicting the efficiency of treatment schemes against *P. falciparum* in Northwest Ecuador (Fig. 1).

Results

Resistance genotypes (2019–2021)

Sixty-seven *P. falciparum* samples were analyzed by Sanger sequencing of the genes: *Pfcrt* (codons 72–76), *Pfdhps* (codons 436, 437, 540, 581 and 613), *Pfdhfr* (codons 50, 51, 59, 108 and 164), *Pfmdr-1* (codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246) and *Pfk13* (codons 476, 493, 539, 543 and 580) to identify SNPs and determine drug resistance genotypes. Samples of local Ecuadorian cases corresponded to the counties of Esmeraldas (n = 27), San Lorenzo (n = 5) and Tobar Donoso (n = 18). Samples obtained from Colombia (n = 17) corresponded to migrants



Fig. 1. Origin of studied isolates. Samples collected from patients in northwest Ecuador: Esmeraldas (n = 27), San Lorenzo (n = 5), Tobar Donoso (n = 18) and Colombia (n = 17). Map created using QGIS software (QGIS 3.34 Prizren) (https://www.qgis.org) and edited with Adobe Illustrator version 27 (https://www.adobe.org).

from Nariño, who were diagnosed in San Lorenzo, Ecuador. The number of samples reflected proportionally the number of reported cases in each location.

Not all the samples amplified or presented results for each of the markers due to low parasitemia or problems in the amplification of each specific marker. The PfCRT haplotypes found in 63 samples were: 61.90% CVM**ET** (39/63), 36.51% CVMN**T** (23/63) and 1.59% CV**IET** (1/63). CVM**ET** was dominant in Esmeraldas in 100% of samples (25/25). In San Lorenzo, the CVM**ET** proportion was 60% (3/5) and was higher than CVMN**T** with 40% (2/5). In Tobar Donoso, CVMN**T** had the highest proportion with 89% (16/18), followed by CVM**ET** with 11% (2/18). In samples obtained from Colombia, 60% were CVM**ET** (9/15) and was the most prevalent haplotype followed by CVMN**T** with 33% (5/15) and CV**IET** with 7% (1/15) (Fig. 2a). The CVM**ET**, CVMN**T** and CV**IET** haplotypes contain the M74**I**, N75**E** and K76**T** mutations associated with CQ resistance[23].

The PfDHPS haplotypes found in 52 samples were: 98.08% of the wild-type SAKAA (51/52) and 1.92% S**G**KAA (1/52), being this last a potentially resistant haplotype. SAKAA was dominant in 100% of Ecuadorian samples: Esmeraldas (24/24), San Lorenzo (5/5) and Tobar Donoso (17/17). While in samples obtained from Colombia, SAKAA was prevalent in 83% of samples (5/6) followed by S**G**KAA in 17% (1/6) (Fig. 2b). In addition, a synonymous mutation (AA**G**) at codon 540 was found in a sample from Esmeraldas and Tobar Donoso and two samples from Colombia. The S**G**KAA haplotype contains the A437**G** mutation associated with resistance to sulfadoxine[32].

The PfDHFR haplotypes found in 63 samples were: 63.49% C**IC**N**I** (40/63), a resistant haplotype, the wild-type 34.92% CNCSI (22/63), and 1.59% CNC**N**I (1/63), a potentially resistant haplotype[29]. C**IC**N**I** was dominant in 100% of Esmeraldas samples (27/27). In San Lorenzo, the C**IC**N**I** proportion was 60% (3/5), and 40% of CNCSI (2/5). All Tobar Donoso samples presented the wild-type, CNCSI (18/18). In samples obtained from Colombia, C**IC**N**I** was present in 77% of samples (10/13) followed by CNCSI in 15% (2/13) and CNC**N**I in 8% of samples (1/13) (Fig. 2c). The C**IC**N**I** and CNC**N**I haplotypes contain the N51**I** and S108**N** mutations associated with pyrimethamine resistance[29].

The PfMDR-1 haplotypes found in 58 samples were: 53.45% NED**FSD**FD (31/58) and 46.55% NED**FSDFY** (27/58). NED**FSD**FD proportion was higher in Tobar Donoso: 100% (18/18) than Esmeraldas: 13% (3/23), while NED**FSDFY** was predominant in 87% of Esmeraldas samples (20/23). In San Lorenzo, NED**FSD**FD was the most prevalent in 60% of samples (3/5) followed by NED**FSDFY** in 40% of samples (2/5). In samples obtained from Colombia, 58% (7/12) had the NED**FSD**FD haplotype and 42% (5/12)had the haplotype NED**FSDFY** (Fig. 2d).

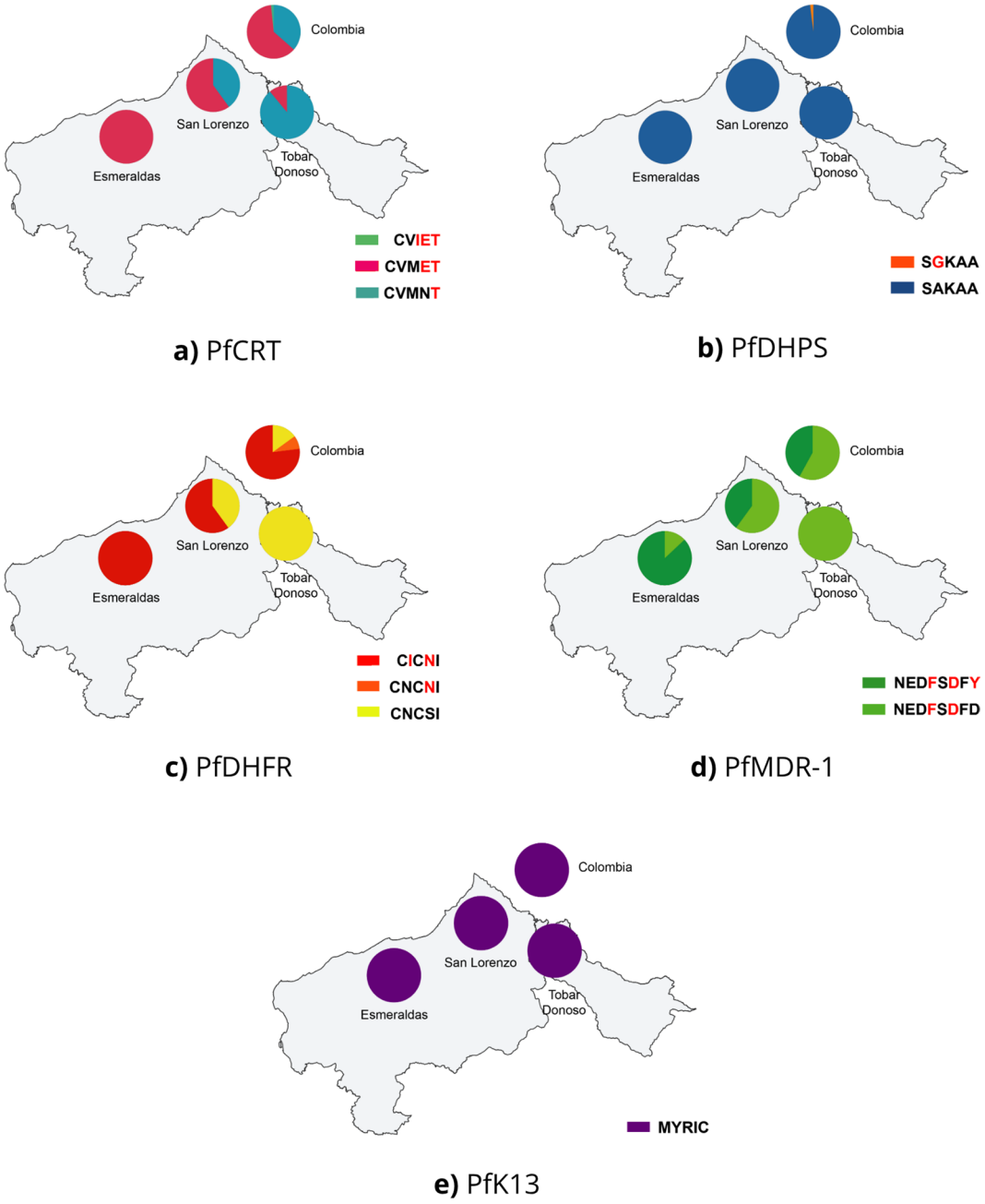


Fig. 2. Geographic distribution of frequencies of drug resistance haplotypes in the 2019–2021 period. (**a**) Haplotypes of PfCRT (72–76), (**b**) Haplotypes of PfDHPS (436, 437, 549, 581 and 613), (**c**) Haplotypes of PfDHFR (50, 51, 59, 108 and 164), (**d**) Haplotypes of PfMDR-1 (codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246), (**e**) Haplotypes of PfK13 (codons 476, 493, 539, 543 and 580).

The PfK13 haplotype found in 52 samples was: 100% the wild-type MYRIC (52/52). MYRIC was present in all samples from Esmeraldas (17/17), San Lorenzo (5/5), Tobar Donoso (18/18) and samples obtained from Colombia (12/12) (Fig. 2e).

Whole-genome sequencing (2019–2021)

Plasmodium falciparum samples underwent whole-genome sequencing (WGS) following selective whole genome amplification (SWGA) to boost the representation of *P. falciparum* DNA within total DNA extracts[55]. WGS could be performed on fifty-four of the sixty-seven *P. falciparum* samples.

WGS coverage varied across sites and samples but confirmed the mutations found by Sanger sequencing in *Pfcrt* (codons 72–76) in 36 samples, *Pfdhps* (codons 436, 437, 540, 581 and 613) in 37 samples, *Pfdhfr* (codons 50, 51, 59, 108 and 164) in 35 samples and *Pfmdr-1* (codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246) in 34 samples.

In addition, WGS found other mutations that were not identified by Sanger sequencing in *Pfcrt* (codons 97, 220, 326, 334, 356 and 371), *Pfdhps* (codon 142), *Pfmdr-1* (codon 35), *Pfk13* (codons 189 and 344) and *Pfaat1* (codons 80, 231 and 258). These mutations are detailed in Table 1.

We found the A16**V** PfDHFR mutation in two samples from Tobar Donoso that have a clonal relationship (genome-wide identity by descent (IBD) > 0.99). This clone was not found within a set of 219 previously sequenced *P. falciparum* genomes that were sampled from the Pacific Coast from 1999 to 201[49]. However, it does have high relatedness (IBD = 0.71) to clonal cluster G from Carrasquilla et al.[49], which also carried the A16**V** PfDHFR variant. We examined the IBD tracts shared between these genomes and found that our new isolates and cluster G are IBD across a 0.8 Mb segment of chromosome 4, which includes the DHFR coding region.

Comparative frequencies of drug resistance haplotypes over time

The spatial and temporal comparison allowed us to characterize the fluctuation of haplotype frequencies between the 2013–2015 and 2019–2021 periods in northwest Ecuador using a Fisher's exact test with a Bonferroni correction to account for the number of comparisons. In PfCRT, CVM**ET** frequency increased in San Lorenzo (from 22 to 60%; not significant (n.s.), significantly increased in Esmeraldas (from 3 to 100%; $P < 0.002$) and appeared in Tobar Donoso (from 0 to 11%; n.s.). On the other hand, CVMN**T** frequency decreased in San Lorenzo (from 78 to 40%; n.s.), in Tobar Donoso (from 1 to 89%; n.s.) and disappeared in Esmeraldas (from 97 to 0%; $P < 0.002$) (Fig. 3a). In PfDHPS, SAKAA frequency remained high in San Lorenzo, Esmeraldas and Tobar Donoso (Fig. 3b). In PfDHFR, the wild-type CNCSI frequency decreased in San Lorenzo (from 83 to 40%; n.s.) and disappeared in Esmeraldas (from 97 to 0%; $P < 0.002$), but it remained high in Tobar Donoso (n.s.). CNC**N**I disappeared in San Lorenzo (from 13 to 0%; n.s.), while C**I**C**N**I frequency significantly increased in Esmeraldas (from 3 to 100%; $P < 0.002$) and increased in San Lorenzo (from 4 to 60%; n.s.) (Fig. 3c). In PfMDR-1, NED**F**S**D**FD frequency significantly decreased in Esmeraldas (from 97 to 13%; $P < 0.002$) and decreased in San Lorenzo (from 78 to 60%; n.s), but it remained high in Tobar Donoso (n.s.). While NED**FSD**F**Y** appeared in Esmeraldas (from 0 to 87%; $P < 0.002$) and San Lorenzo (from 0 to 40%; n.s.). Surprisingly, haplotypes with single mutations NED**F**SNFD (Y184**F**) and NEDYS**D**FD (N1042**D**) found in the 2013–2015 period disappeared in San Lorenzo (from 22 to 0%) and Esmeraldas (from 3 to 0%), although these changes were not statistically significant (Fig. 3d; n.s.). Finally, in PfK13, the wild-type MYRIC frequency has remained high in San Lorenzo, Esmeraldas and Tobar Donoso (Fig. 3e).

Genetic diversity of drug resistance genes

The nucleotide diversity (π) and haplotype diversity (H_d) were calculated to determine the genetic diversity of *Pfcrt*, *Pfdhps*, *Pfdhfr* and *Pfmdr-1* in *P. falciparum* from northwest Ecuador and samples obtained from Colombia. The nucleotide diversity (π) measured the degree of DNA polymorphism, while haplotype diversity (H_d) measured the singularity of a haplotype in relation to sample size[56]. At *Pfcrt*, samples obtained from Colombia ($\pi = 0.080$, $H_d = 0.562$) and San Lorenzo ($\pi = 0.080$, $H_d = 0.600$) showed higher diversity than Tobar Donoso ($\pi = 0.028$, Hd = 0.209) and no diversity was found in Esmeraldas. At *Pfdhps*, samples obtained from Colombia ($\pi = 0.080$, $H_d = 0.600$) had higher diversity than Tobar Donoso ($\pi = 0.008$, $H_d = 0.118$) and Esmeraldas ($\pi = 0.005$, $H_d = 0.080$) and no diversity was found in San Lorenzo. At *Pfdhfr*, San Lorenzo ($\pi = 0.080$, $H_d = 0.600$) had higher diversity than Colombia ($\pi = 0.040$, $H_d = 0.410$) and no diversity was found in Esmeraldas and Tobar Donoso. At *Pfmdr-1,* samples obtained from Colombia ($\pi = 0.022$, $H_d = 0.530$) and San Lorenzo ($\pi = 0.025$, $H_d = 0.600$) had higher diversity than Esmeraldas ($\pi = 0.009$, $H_d = 0.220$) and no diversity was found in Tobar Donoso (Table 2). *Pfk13* was excluded from the analysis because it did not exhibit polymorphisms in the molecular markers analyzed.

Gene	N	Percentage (%)	Coding variant
Pfcrt	19/36	52.78	H97**Q**
	36/36	100	A220**S**
	15/37	40.54	N326**D**
	15/37	40.54	S334**N**
	16/38	42.11	I356**L**
	21/38	55.26	R371**T**
Pfdhps	16/36	44.44	K142**K**(AA**A**)
Pfdhfr	2/30	6.67	A16**V**
Pfmdr-1	2/23	8.70	I35**K**
Pfk13	19/19	100	K189**T**
	2/19	10.53	A344**V**
Pfaat1	7/7	100	D80**E**
	19/33	57.58	V231**D**
	17/35	48.57	S258**L**

Table 1. Frequencies of coding variants found in drug resistance genes obtained by WGS. *n* number of samples, *SNP* single nucleotide polymorphism.

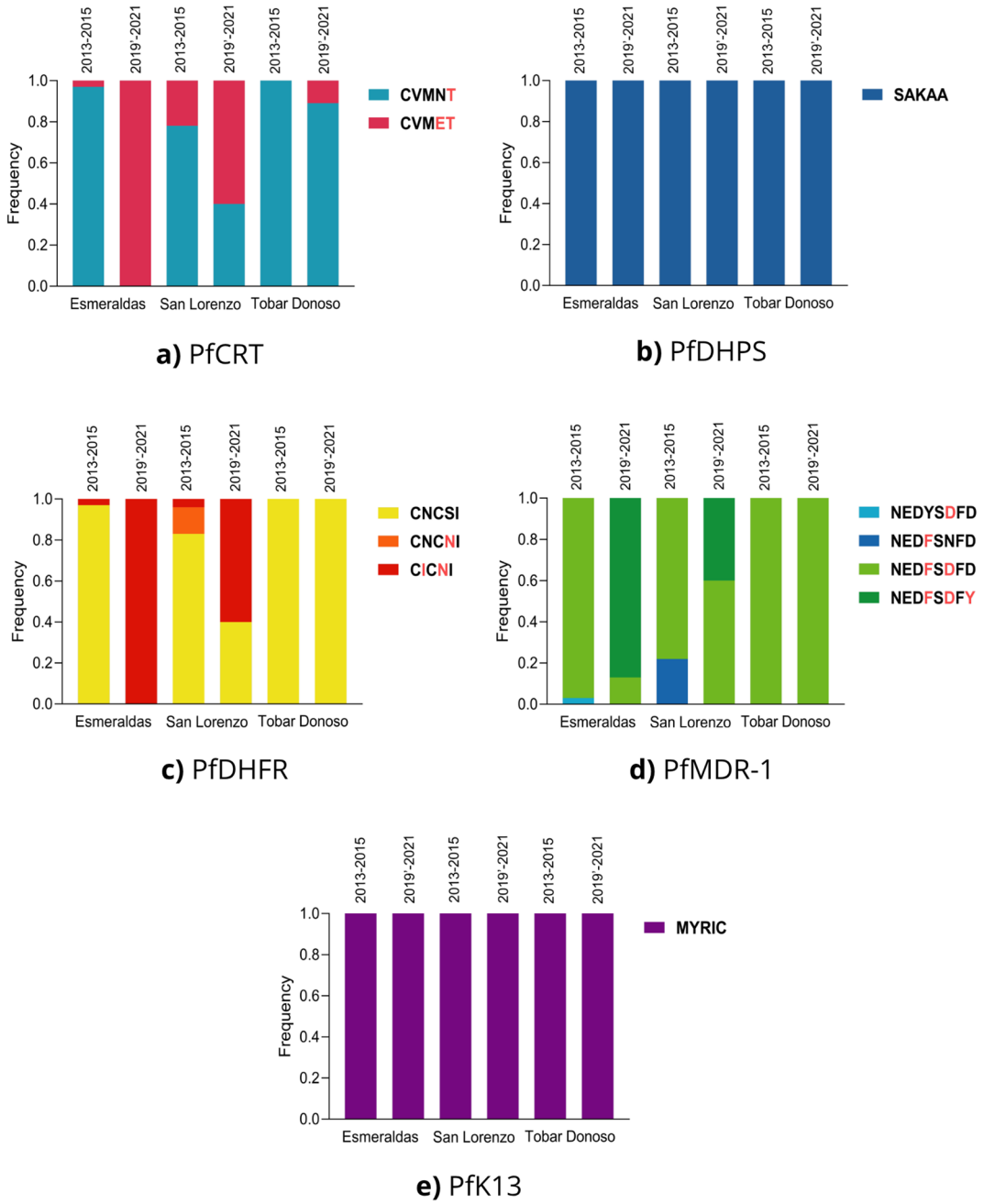


Fig. 3. Comparative frequencies of drug resistance haplotypes in Northwest Ecuador between the 2013–2015 and 2019–2022 periods. (**a**) Haplotypes of PfCRT (72–76), (**b**) Haplotypes of PfDHPS (436, 437, 549, 581 and 613), (**c**) Haplotypes of PfDHFR (50, 51, 59, 108 and 164), (**d**) Haplotypes of PfMDR-1 (codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246), (**e**) Haplotypes of PfK13 (codons 476, 493, 539, 543 and 580).

Discussion

The efficacy of antimalarials has contributed to the reduction of morbidity and mortality rates of malaria[57]. However, the spread and increase of antimalarial resistance contribute to the emergence of malaria outbreaks and hinder malaria elimination[24,42]. This study focused on the molecular epidemiology of drug resistance genes (*Pfcrt*, *Pfdhps*, *Pfdhfr*, *Pfmdr-1*, *Pfk13* and *Pfaat1*) in *P. falciparum* in northwestern Ecuador between 2019 and 2021. We show *P. falciparum* CQ and pyrimethamine resistance expansion and most importantly the appearance of a mutation associated with proguanil resistance for the first time in Ecuador.

The haplotypes found in *Pfcrt* (Fig. 2a) are clear evidence of the strong pressure that CQ has exerted over *P. falciparum* for more than 5 decades in Ecuador (Fig. 4)[49,58]. It is evident that in Ecuador as in the rest of South America[20,21,26], CQ resistance has become fixed as all sampled parasites carried the K76**T** coding mutation in *Pfcrt*. Reversal of CQ resistance has not been seen in this region even though CQ is no longer used against *P. falciparum*[26,59]. However, in French Guiana, reversal of CQ resistance has been acquired by the C350**R** mutation

Gene	Locality	n	S	k	π	h	Hd
Pfcrt	Tobar Donoso	18	2	0.418	0.028	2	0.209
	Esmeraldas	25	0	0.000	0.000	1	0.000
	San Lorenzo	5	2	1.200	0.080	2	0.600
	Colombia	15	3	1.090	0.080	3	0.562
Pfdhps	Tobar Donoso	17	1	0.018	0.008	2	0.118
	Esmeraldas	24	1	0.080	0.005	2	0.080
	San Lorenzo	5	0	0.000	0.000	1	0.000
	Colombia	6	3	1.200	0.080	3	0.600
Pfdhfr	Tobar Donoso	18	0	0.000	0.000	1	0.000
	Esmeraldas	27	0	0.000	0.000	1	0.000
	San Lorenzo	5	2	1.200	0.080	2	0.600
	Colombia	13	2	0.670	0.040	3	0.410
Pfmdr-1	Tobar Donoso	18	0	0.000	0.000	1	0.000
	Esmeraldas	25	1	0.220	0.009	2	0.220
	San Lorenzo	5	1	0.600	0.025	2	0.600
	Colombia	12	1	0.530	0.022	2	0.530

Table 2. Genetic diversity of molecular markers of drug resistance genes in *P. falciparum*. n: Number of samples, S: Number of polymorphic sites, k: Average number of pairwise nucleotide differences, π: Nucleotide diversity, h: Number of haplotypes, *Hd* haplotype diversity.

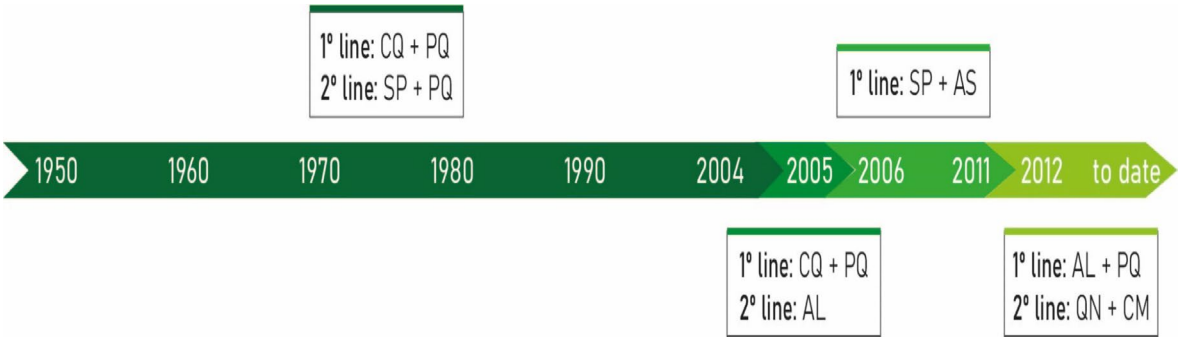


Fig. 4. Antimalarial treatments against *P. falciparum* in Ecuador. Timeline of antimalarial treatments since 1950 according to the Ministry of Public Health of Ecuador. Chloroquine (CQ), primaquine (PQ), sulfadoxine-pyrimethamine (SP), artemether-lumefantrine (AL), artesunate (AS), quinine (QN) and clindamycin (CM). *Source*: Aguilar et al.[58].

in *Pfcrt*[60]. In particular, the indiscriminate use of CQ reported in several areas of the Pacific coast during the 1990s contributed to the selection of CQR parasites[61]. Besides, clonal expansions, may have caused the fast spread and increase of CQR parasites[21], as well as the loss of CQ-sensitive parasites. Specifically, in Ecuador CQ is not used against *P. falciparum* since 2006 and CQ has not been used in neighboring countries since the beginning of the century.

Other mutations such as H97**Q**, A220**S**, Q271**E,** N326**D**, C350**S**, I356**T/L** and R371**I** in PfCRT have been associated with CQR parasites[16,17]. In this study, H97**Q**, A220**S**, N326**D**, S334**N**, I356**L** and R371**T** were found (Table 1). The presence of these mutant alleles may have contributed to CQ resistance in Ecuadorian *P. falciparum* because it has been suggested that at least three complementary mutations to K76**T** are necessary to confer high degrees of CQ resistance[18].

Resistance mutations in *Pfdhfr* were found in 65% of Ecuadorian samples (Fig. 2c). In contrast, sulfadoxine resistance appears low as only one instance of a mutant PfDHPS haplotype, S**G**KAA, (Fig. 2b) was found among the parasites originating from Colombia, but not Ecuador. This difference may stem from the history of drug use in the region. In South America, pyrimethamine was used massively as a prophylactic against malaria in the 1950s[33]. Later, in the 1970s, pyrimethamine was co-administrated with sulfadoxine to combat CQR parasites[14]. Consequently, SP resistance has been reported in South America since 1980[26]. Specifically, in Ecuador, SP was implemented as a second-line treatment in 1950 and, in 2005, it was used as a first-line treatment until 2006 (Fig. 4)[58]. Therefore, the emergence of resistant parasites was caused by the pressure exerted by SP for more than 5 decades. However, the intermittent use of SP temporarily reduced SP pressure, which may have caused SP-sensitive parasites to continue to circulate in the region.

Importantly, the genetic monitoring of *Pfdhfr* further suggests that resistance to proguanil has appeared in Ecuador. *Plasmodium falciparum* acquires resistance to cycloguanil, the active metabolite of proguanil via combined coding mutations for A16**V** and S108**N/T** in PfDHFR[14,28,62]. In Colombia, A16**V** was detected for the first time in combination with CNCSI and C**I**C**N**I in Nariño between 2012 and 2013[31]. In this study, A16**V** with CNCSI was identified in two samples from Tobar Donoso (Table 1) and was previously detected in isolates from San Lorenzo[49]. While A16**V** was reported in Pacific coast parasites, the samples from San Lorenzo and Tobar Donoso are not the same clones. Instead, our IBD analysis shows that A16V was passaged onto a new genomic background through recombination[49]. Atovaquone-proguanil has been commercialized in Nariño, Colombia for many years[31], suggesting that the emergence of A16**V** in Colombia has been caused by the use of atovaquone-proguanil as prophylaxis against malaria[31]. So, the presence of A16**V** in Tobar Donoso could be explained by the recent recombination of parasites in the Pacific Coast of Colombia or Ecuador[49]. Given the temporal order of sampling, the Tobar Donoso genomes are likely the progeny of a cross between clonal cluster G (Carrasquilla et al.[49]), which carries the A16**V** allele, and another parasite, but we were unable to identify the second putative parent within our extended Pacific Coast data set. The possible spread of this mutation in Ecuador and Colombia raises an important point against the future use of proguanil in the region.

At PfMDR-1, a shift in haplotype frequencies across time was found (Fig. 3d). While both common haplotypes have been circulating in South America for many years, the most frequent haplotype in 2019–2021 (the triple mutant NED**FS**D**F**Y) has not been sampled previously in Ecuador. Mutations in *Pfmdr-1* have an important role in response to different antimalarials such as CQ, QN, AQ, MQ and LF[6,14,15]. However, South American *P. falciparum* isolates have not shown phenotypic changes in drug response even though Y184**F**, N1042**D** and D1246**Y** are commonly present[20,63]. An in vitro assay tested parasites from Nariño with NED**FS**D**F**D and NED**FS**D**F**Y haplotypes, which were shown to be sensitive to AQ, MQ and LF[63]. In Esmeraldas, ESM-2013 (isolated from 2013) with Y184**F** and N1042**D** mutation showed sensitivity to QN, LF and AR[20]. Mutations in other genes located in the DV such as *Pfcrt* and *Pfaat1* may therefore influence resistance to these drugs[11,64].

The prevalence of MYRIC haplotype (wild-type) in *Pfk13* (Fig. 2e) shows sensitivity to ARTs in 100% of Ecuadorian *P. falciparum* samples. Previous studies in Ecuador, Colombia, Peru, Venezuela, Brazil, Suriname and French Guiana have not found any mutation in *Pfk13* associated with ART resistance[20,42,43]. However, in Guyana, the C580**Y** mutation was detected in 2010[41,42]. There is, therefore, concern that C580**Y** could spread to other regions of South America, and molecular studies are key to monitoring for any spread or new emergence. In addition, K189**T** mutation in PfK13, not associated with ART resistance, was identified in this study and recently reported in Ecuador, Colombia, Venezuela, Brazil, Guyana and French Guiana[42,49]. In Ecuador, AS was used with SP as a first-line treatment in 2006; however, AL was already used as a second-line treatment in 2005[58]. AL has been the official first-line treatment against *P. falciparum* since 2012 (Fig. 4)[58].

In the last decade, resistance to quinolines such as CQ, QN, AQ and MQ has been associated with mutations in *Pfaat1*[44–46]. The S258**L** mutation identified in this study (Table 1) shows evidence of having been under selection in the Pacific coast of Ecuador and Colombia[49]. Recently, Amambua-Ngwa et al.[48] found the same mutation segregating in Gambian parasites, and through CRISPR/Cas9 editing, validated its role in CQ resistance. Likewise, an in vitro study showed resistance to three compounds when P380**S**, K238**N**, V185**L**, F230**L** mutations and an insertion at codon 35 were present in *Pfaat1*[47]. This pressure could have been caused mostly by CQ and QN, used historically in South America[26,65], and AQ which was introduced in several areas of South America, including the Pacific coast, since 1949[24,59,66].

The observed pattern of resistance does not match the current drug use as has been observed elsewhere in South America, Instead, the recent increase in proportions of mutant haplotypes in Ecuador such as CVM**ET** (double mutant) in PfCRT, C**I**C**N**I (double mutant) in PfDHFR and the appearance of NED**FS**D**F**Y (triple mutant) in PfMDR-1 (Fig. 3) could be explained by clonal expansions caused by parasites from migrant humans and not by the pressure resulting from CQ and SP treatments, because they have not been used against *P. falciparum* in the last 2 decades. A previous analysis of neutral microsatellites found that outbreaks that occurred in Tobar Donoso and Esmeraldas in 2019 and 2020, respectively, were mostly caused by a single parasite clone[67]. Furthermore, CVM**ET** in PfCRT, C**I**C**N**I in PfDHFR and NED**FS**D**F**Y in PfMDR-1 were dominant in samples from migrant human hosts in this study (Fig. 2), and they have been found circulating in Nariño since 1999[24,26,31]. While in northwest Ecuador, CVM**ET** and C**I**C**N**I were reported in low proportions in 2013–2015[20], and NED**FS**D**F**Y was absent, but they are now common in recent samples in this study. This suggests that recent outbreaks that occurred in Ecuador may have resulted from the spread of parasites by migrant humans from Nariño. In agreement, the findings of this study show that genetic diversity (both nucleotide and haplotype) in drug resistance genes (*Pfcrt, Pfdhps, Pfdhfr and Pfmdr-1*) is low. San Lorenzo and Nariño, Colombia presented relatively more genetic variability than other sampled locations in this study. The low diversity in Esmeraldas and Tobar Donoso was mostly due to the fact that samples came from clonal outbreaks in which single haplotypes were predominant. The drug resistance haplotype diversity mirrors diversity seen with neutral microsatellite markers.

Previously, an important human migratory flow that contributes to the spread of parasites on the Pacific coast of Colombia has already been reported[51]. In the present study, the main routes of cross-border and intra-border human migration contributing to the spread of malaria parasites in the North coast of Ecuador and the South coast of Colombia are proposed (Fig. 5). Specifically, the spread of resistant parasites to Ecuador could have occurred in two ways: (1) by Ecuadorians who traveled to Colombia and became infected with parasites from there, and (2) directly by infected Colombians who arrived in Ecuador. The main migration route between Nariño and San Lorenzo for several years has been by their seaports. Therefore, Colombian parasites could have reached the coast of Ecuador by the constant migratory flow along the Pacific coast due to the shellfish trade, legal and illegal mining, agriculture and illegal activities such as drug trafficking through the seaports of Nariño, San Lorenzo and Esmeraldas[68–70]. The opening of a new road between Colombia and Ecuador, and

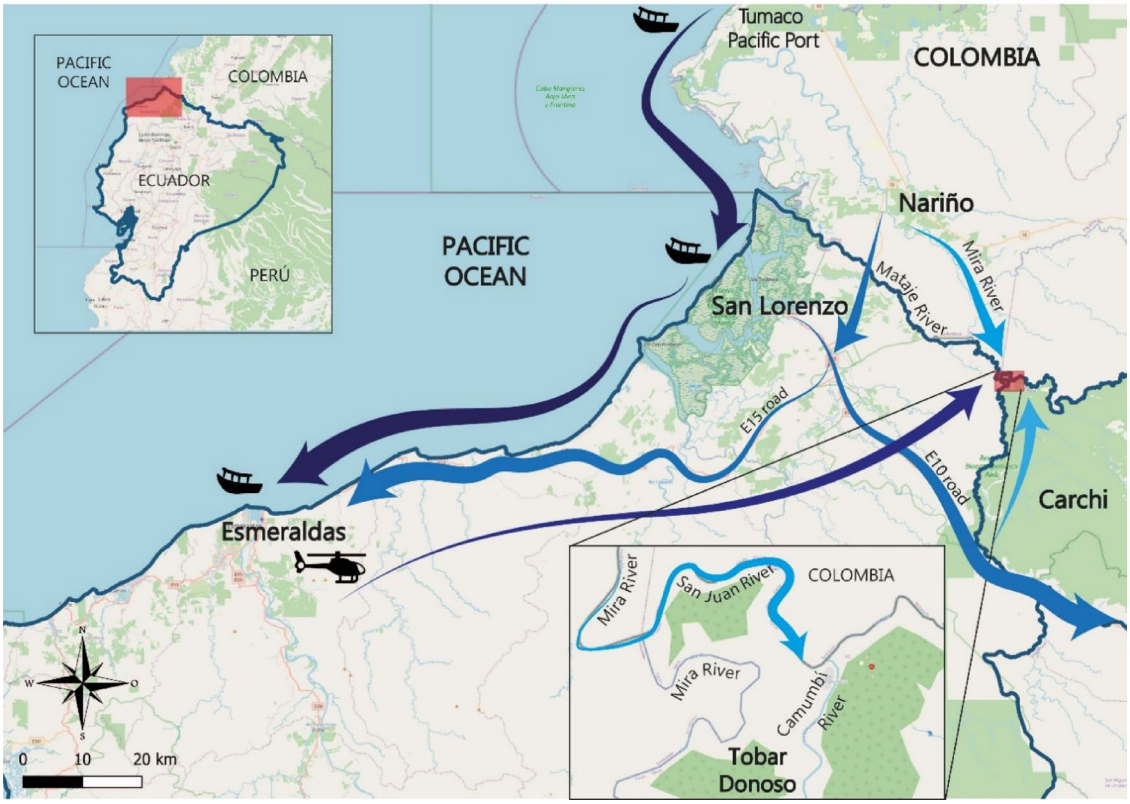


Fig. 5. Cross-border and intra-border routes of entry of *P. falciparum* in northwestern Ecuador. Possible routes of the human migration used for the passage of *P. falciparum* in the south coast of Colombia and the north coast of Ecuador. Elaborated according to the information collected from oral reports. Map created using QGIS software (QGIS 3.34 Prizren) (https://www.qgis.org) and edited with Adobe Illustrator version 27 (https://www.adobe.org).

the bridge over Mataje River that connects Nariño and San Lorenzo[71], could represent an important passage of malaria infected people between Ecuador and Colombia. In addition, the E15 road that connects San Lorenzo and Esmeraldas is an important corridor in the spread of malaria parasites (Fig. 5). The presence of A16**V** in PfDHFR (Table 1) in Tobar Donoso also suggests a process of human migration because this mutation has already found in San Lorenzo[49] and is common in Nariño[31].

Even though, cross border migration maintains transmission in eliminating areas, only a small number of studies have focused on cross border migration that maintains malaria transmission and spreads drug resistance mutations throughout the region. The study by Carrasquilla et al.[49] performed whole-genome sequencing of parasites from Ecuador and Colombia and found high relatedness and shared circulating clones. Previously, a study by Griffing et al.[30] has shown how *P. falciparum* lineages with unique drug resistance mutations repopulated Peru after near elimination in the 1990s. Similarly, a study by Ventocilla et al.[72] suggests population flow between *P. vivax* parasites in Ecuador and Peru. This current study complements these previous findings by suggesting that migratory routes carry new drug resistance mutations to areas where they were not previously reported.

A possible limitation of this study is the low number of samples in some of the localities such as San Lorenzo. The number of samples mirrors the small number of reported cases in those localities and shows that even a small number of parasites circulating can cause important outbreaks that bring large shifts in genotype frequencies and continue spatio-temporal allele fluctuations. Consequently, in a region in pre-elimination with these epidemiological characteristics, genotype predictions will be difficult and continued monitoring is needed.

Factors such as drug pressure and host human migration influence the emergence and spread of resistant parasites which contribute to the appearance of malaria outbreaks and complicate malaria control and elimination in Ecuador. The findings of this work indicate that CQ and pyrimethamine-resistant parasites have increased in recent years, while artemisinin-sensitive parasites prevail in Ecuador. Therefore, this study shows that the current treatment (artemether + lumefantrine) likely remains effective against *P. falciparum* and supports the continuation of its use as a main treatment, but warns against the use of CQ, SP or proguanil. We note, however, that we employ a candidate gene approach and cannot rule out the presence of novel resistance-linked mutations elsewhere in the genome.

In addition, the genotypes of *P. falciparum* show rapid fluctuations within the region, likely due to the boom-bust dynamics of clonal lineages coupled with frequent migration of human hosts across the Ecuador-Colombia border which may make the spread of drug resistant parasites in the region less predictable, increasing the need for continued monitoring of genetic drug resistance profiles in low transmission areas.

Finally, this study demonstrates the importance of the epidemiological characterization of resistance in *P. falciparum* in a region in the process of eliminating malaria: genetic surveillance of drug resistance genes is key to responding in a timely manner in case of therapeutic failure because it informs treatment success or failure before clinical observations to prevent the spread of malaria.

Methods

Study area and parasite population

The study was carried out in Ecuador (South America) from 2019 to 2021, with samples collected in Esmeraldas and Carchi provinces located in the Northwest on the border with Colombia (Fig. 1). Sixty-seven blood samples were collected on Whatman filter paper from patients diagnosed with *P. falciparum* by microscopy or RDTs (Rapid Diagnostic Tests). Informed consent was obtained from all participants and/or their legal guardians. Samples were collected by personnel of the MPH between 2019 and 2021 and delivered to CISeAL (Centro de Investigación para la Salud en América Latina) for molecular analysis. During this period, 702 infections by *P. falciparum* were registered in Northwest Ecuador[4]. Samples of local Ecuadorian cases corresponded to the counties of Esmeraldas (40.29%), San Lorenzo (7.46%) and Tobar Donoso (26.86%). The remaining 25.37% corresponded to migrants from Nariño, Colombia diagnosed in San Lorenzo. The number of samples collected proportionally represents the number of reported samples in the study area. Additionally, data of *P. falciparum* samples (2013–2015 period) from a previous study in Ecuador were included for temporal comparisons[20].

DNA extraction

DNA extraction of *P. falciparum* was performed from blood samples on Whatman filter paper stored at 4 °C, using the commercial Purelink™ Genomic DNA Mini Kit, INVITROGEN following the protocol recommended by the manufacturer (Invitrogen, Life Technology, USA), and stored at − 20 °C for its molecular analysis.

Amplification of drug resistance genes

DNA was amplified for molecular markers of drug resistance genes: *Pfcrt* (codons 72–76), *Pfdhps* (codons 436, 437, 540, 581 and 613), *Pfdhfr* (codons 50, 51, 59, 108 and 164), *Pfmdr-1* (codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246) and *Pfk13* (codons 476, 493, 539, 543 and 580).

Polymerase Chain Reaction (PCR) solution contained 12.5 µl of Colorless GoTaq Master Mix (Promega, Madison, WI, USA), 2.5 µl of $MgCl_2$ (25 mM) (Promega, Madison, WI, USA), 3 µl of ultrapure H_2O, 1 µl of primers forward and reverse (10 µM) and 5 µl of DNA. Nested PCR protocols (Table 3) established in previous studies were used[20]. Amplifications were carried out in SureCycler 8800 (Agilent Technologies, USA).

Amplicon sizes were verified in 2% agarose gel and observed in UVP BioDoc-It digital transilluminator (UVP, CA, UK). After confirmation, amplified DNA was sent for purification and Sanger sequencing to Macrogen, South Korea.

Determination of genotypes in drug resistance genes

The NF54 strain of *P. falciparum* was used as a control to identify SNPs (Single Nucleotide Polymorphisms) in *Pfcrt* (codons 72–76), *Pfdhps* (codons 436, 437, 540, 581 and 613), *Pfdhfr* (codons 50, 51, 59, 108 and 164), *Pfmdr-1* (codons 86, 130, 144, 184, 1034, 1042, 1226 and 1246) and *Pfk13* (codons 476, 493, 539, 543 and 580). All sequences were analyzed and aligned in Geneious software version 2020.0.1 (Biomatters, Inc, Newark, NJ, USA). Sanger sequences of drug resistance genes were used for genetic diversity calculation.

Whole-genome sequencing

Fifty-four DNA samples underwent selective whole genome amplification (SWGA) to boost the representation of *P. falciparum* DNA within total DNA extracts[55]. The Broad Institute Genomics Platform then constructed NEBNext libraries and performed 150-bp paired-end sequencing on an Illumina NovaSeqSP instrument. Sequences were aligned and variants were called using a standard BWA-GATK3 pipeline following the best practices established by the Pf3k consortium[76]. The WGS results were used to confirm the genotypes obtained by Sanger sequencing, to identify variants in the same genes outside the Sanger-sequenced regions and to identify variants in *Pfaat1*, that were not previously sequenced in Ecuador by Sanger sequencing.

Data analysis

GraphPad Prism version 8.0.2 was used to calculate frequencies and make comparative graphs of *Pfcrt*, *Pfdhps*, *Pfdhfr*, *Pfmdr-1* and *Pfk13* haplotypes in each locality from the 2013–2015 and 2019–2022 periods. To compare the frequencies of haplotypes between two different time points, a Fisher's Exact test was used with a Bonferroni correction for 20 comparisons. Significance was defined as $p < 0.05$. The number of polymorphic sites (S), the average number of pairwise nucleotide differences (k), nucleotide diversity (π), number of haplotypes (h) and haplotype diversity (Hd) were calculated in DnaSP software version 6.12.03.

Relatedness analysis

We estimated relatedness between all genome pairs using the program hmmIBD[77]. The algorithm estimates fractional (genome-wide) relatedness as identity by descent (IBD) and provides IBD status at all genomic coordinates included in the analysis. We used the length and position of these IBD tracts to infer relatedness at the *Pfdhfr* locus. The IBD analysis included both the Ecuador samples from this study and 219 additional samples from the South American Pacific Coast originally analyzed in Carrasquilla et al.[49]. We excluded from the IBD analysis any genome with less than 30% of the genome covered at 5x.

Gene	Primers 5′–3′	Amplicon	Reference
Pfcrt	**F1:**AGCAAAAATGACGAGCGTTATAG	559 pb	Griffing et al.[30]
	R1:ATTGGTAGGTGGAATAGATTCTC		
	F2:TTTTTCCCTTGTCGACCTTAAC	264 pb	
	R2:AGGAATAAACAATAAAGAACATA ATCATAC		
Pfdhfr	**F1:**TCCTTTTTATGATGGAACAAG	633 pb	Alam et al.[73]
	R1:AGTATATACATCGCTAACAGA		
	F2:TTTATGATGGAACAAGTCTGC	616 pb	
	R2:ACTCATTTTCATTTATTTCTGG		
Pfdhps	**F2:**GATTCTTTTTCAGATGGAAA	770 pb	Fierro[31]
	R2:TTCCTCATGTAATTCATCTGA		
	F1:AACCTAAACGTGCTGTTCAA	711 pb	
	R1:AATTGTGTGATTTGTCCACAA		
Pfmdr-1 (86)	**F1:**CCGTTTAAATGTTTACCTGCAC	1014 pb	Vinayak et al.[74]
	R1:TGGGGTATTGATTCGTTGCAC		
	F2:GTATGTGCTGTATTATCAGGAG	746 pb	
	R2:AGCCTCTTCTATAATGGACATG		
Pfmdr-1 (1034)	**F1:**TTTAAAGATCCAAGTTTTTTAAT ACAGGAAGC	924 pb	Montenegro et al.[63]
	R1:CTTACTAACACGTTTAACATCTTC CAATG		
	F2:AAGCTATTGATTATAAAAATAAAG GACAAAAAAGAAG	761 pb	
	R2:GGACATATTAAATAACATGGGTT CTTGAC		
Pfk13	**F1:**GCAAATAGTATCTCGAAT	2126 pb	Talundzic et al.[75]
	R1:CTGGGAACTAATAAAGAT		
	F2:GATAAACAAGGAAGAATATTCT	748 pb	
	R2:CGGAATCTAATATGTTATGTTCA		

Table 3. Primers used for the nested PCR for molecular markers of drug resistance genes in *P. falciparum*. *Pfcrt*: *Plasmodium falciparum* chloroquine resistance transporter; *Pfdhfr*: *Plasmodium falciparum* dihydropteroate synthetase; *Pfdhps*: *Plasmodium falciparum* dihydrofolate reductase; *Pfmdr-1*: *Plasmodium falciparum* multidrug resistance 1; *Pfk13*: *Plasmodium falciparum* kelch 13.

Data availability

Data analyzed in this study is available in CISeAL and can be requested to Dr. Fabián Sáenz. All whole-genome sequence data are available on the NCBI Sequence Read Archive as BioProject PRJNA759192.

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Author contributions
IÑ performed the experiments; IÑ and FES designed the study. IÑ, FES, AME and DEN wrote the manuscript; AME and DEN obtained WGS data, JB performed field work.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
All methods were performed in accordance with guidelines and regulations and in accordance with the declaration of Helsinki. The study was approved by the Research Ethics Committee on Human Beings of Pontificia Universidad Católica del Ecuador registered in the document CEISH-571-2018 and by the Ministry of Public Health (MPH) of Ecuador registered in the document MSP-DIS-2019-044-O.

Additional information
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